BASES DE DATOS

BD02

NEREA MONTOYA OLIVARES

EXERCISE 1

We are going to create some useful tables for an Estate Agency. We will start by
creating the following tables with SQL using the following information:

- The ***CLIENT*** table includes information about the clients: client identifier, identification card, name, surnames, register date, email, mobile phone, nationality.

- The ***PROPERTY*** table includes information about the properties: reference (number that identifies the property), location, price, type, number of bedrooms, number of bathrooms, build area, plot size, features (long description about garden, garage, sea view, community pool, parking, …), for sale (true / false).

- The ***VISIT*** table includes information about scheduled visits of clients to the properties: visit identifier, visit date and hour (only one field for date and hour), agent name, property, client identifier.

Consider the following:

a. Choose the most adequate attribute types depending on their content.

b. Establish the following constraints:

1. We can distinguish rows in the tables by the identifiers or references.

2. *Email* and *mobile phone* can not be repeated.

3. The *property* of the VISIT must be a reference to the PROPERTY table.

4. The *client* of the VISIT can only be a *client identifier* from the CLIENT table.

5. The only information that can be empty is the *plot size*.

6. *Build area* must be lesser than *plot size*.

7. The *type* attribute can only take these values: Commercial, Finca, Land, Penthouse, Townhouse, Villa.

8. The *sale* default value is false, and the default value for *register date* is the current date.

SCRIPT:

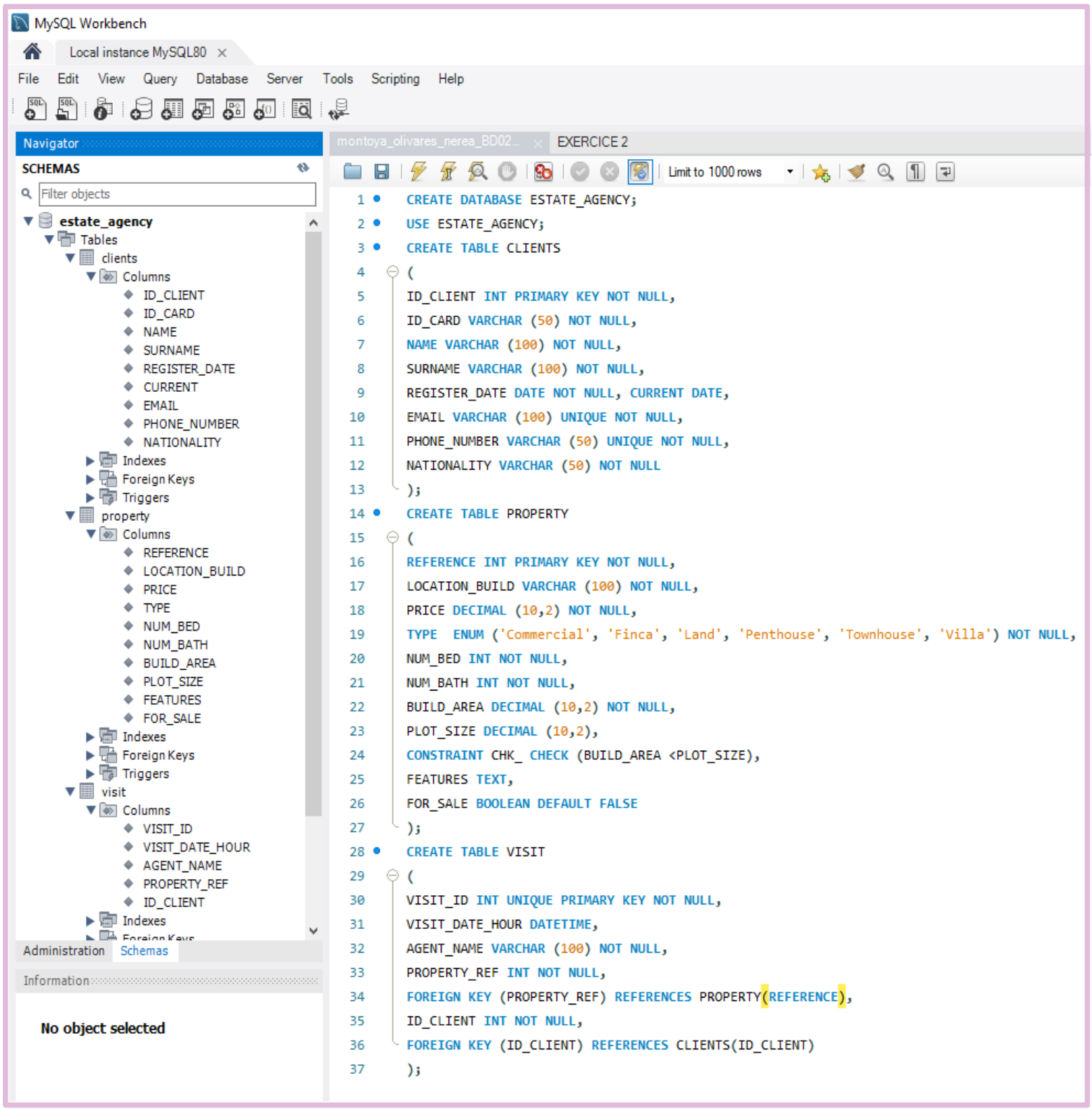
```
CREATE DATABASE ESTATE_AGENCY;

USE ESTATE_AGENCY;

CREATE TABLE CLIENTS
(
ID_CLIENT INT PRIMARY KEY NOT NULL,
ID_CARD VARCHAR (50) NOT NULL,
NAME VARCHAR (100) NOT NULL,
SURNAME VARCHAR (100) NOT NULL,
REGISTER_DATE DATE NOT NULL, CURRENT DATE,
EMAIL VARCHAR (100) UNIQUE NOT NULL,
PHONE_NUMBER VARCHAR (50) UNIQUE NOT NULL,
NATIONALITY VARCHAR (50) NOT NULL
);

CREATE TABLE PROPERTY
(
REFERENCE INT PRIMARY KEY NOT NULL,
LOCATION_BUILD VARCHAR (100) NOT NULL,
PRICE DECIMAL (10,2) NOT NULL,
TYPE  ENUM ('Commercial', 'Finca', 'Land', 'Penthouse', 'Townhouse',
'Villa') NOT NULL,
NUM_BED INT NOT NULL,
NUM_BATH INT NOT NULL,
BUILD_AREA DECIMAL (10,2) NOT NULL,
PLOT_SIZE DECIMAL (10,2),
CONSTRAINT CHK_ CHECK (BUILD_AREA <PLOT_SIZE),
FEATURES TEXT,
FOR_SALE BOOLEAN DEFAULT FALSE
);

CREATE TABLE VISIT
(
VISIT_ID INT UNIQUE PRIMARY KEY NOT NULL,
VISIT_DATE_HOUR DATETIME,
AGENT_NAME VARCHAR (100) NOT NULL,
PROPERTY_REF INT NOT NULL,
FOREIGN KEY (PROPERTY_REF) REFERENCES PROPERTY(REFERENCE),
ID_CLIENT INT NOT NULL,
FOREIGN KEY (ID_CLIENT) REFERENCES CLIENTS(ID_CLIENT)
);
```


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
estate_agency
Tables
clients
Columns
ID_CLIENT
ID_CARD
NAME
SURNAME
REGISTER_DATE
CURRENT
EMAIL
PHONE_NUMBER
NATIONALITY
Indexes
Foreign Keys
Triggers
property
Columns
REFERENCE
LOCATION_BUILD
PRICE
TYPE
NUM_BED
NUM_BATH
BUILD_AREA
PLOT_SIZE
FEATURES
FOR_SALE
Indexes
Foreign Keys
Triggers
visit
Columns
VISIT_ID
VISIT_DATE_HOUR
AGENT_NAME
PROPERTY_REF
ID_CLIENT
Indexes
Administration Schemas
Information
No object selected
montoya_olivares_nerea_BD02...
EXERCICE 2
Limit to 1000 rows
1 CREATE DATABASE ESTATE_AGENCY;
2 USE ESTATE_AGENCY;
3 CREATE TABLE CLIENTS
4 (
5 ID_CLIENT INT PRIMARY KEY NOT NULL,
6 ID_CARD VARCHAR (50) NOT NULL,
7 NAME VARCHAR (100) NOT NULL,
8 SURNAME VARCHAR (100) NOT NULL,
9 REGISTER_DATE DATE NOT NULL, CURRENT DATE,
10 EMAIL VARCHAR (100) UNIQUE NOT NULL,
11 PHONE_NUMBER VARCHAR (50) UNIQUE NOT NULL,
12 NATIONALITY VARCHAR (50) NOT NULL
13);
14 CREATE TABLE PROPERTY
15 (
16 REFERENCE INT PRIMARY KEY NOT NULL,
17 LOCATION_BUILD VARCHAR (100) NOT NULL,
18 PRICE DECIMAL (10,2) NOT NULL,
19 TYPE ENUM ('Commercial', 'Finca', 'Land', 'Penthouse', 'Townhouse', 'Villa') NOT NULL,
20 NUM_BED INT NOT NULL,
21 NUM_BATH INT NOT NULL,
22 BUILD_AREA DECIMAL (10,2) NOT NULL,
23 PLOT_SIZE DECIMAL (10,2),
24 CONSTRAINT CHK_ CHECK (BUILD_AREA <PLOT_SIZE),
25 FEATURES TEXT,
26 FOR_SALE BOOLEAN DEFAULT FALSE
27);
28 CREATE TABLE VISIT
29 (
30 VISIT_ID INT UNIQUE PRIMARY KEY NOT NULL,
31 VISIT_DATE_HOUR DATETIME,
32 AGENT_NAME VARCHAR (100) NOT NULL,
33 PROPERTY_REF INT NOT NULL,
34 FOREIGN KEY (PROPERTY_REF) REFERENCES PROPERTY(REFERENCE),
35 ID_CLIENT INT NOT NULL,
36 FOREIGN KEY (ID_CLIENT) REFERENCES CLIENTS(ID_CLIENT)
37);

EXERCISE 2

We are going to modify the tables created in the first exercise.

1. Create a *pool* field in the PROPERTY table. Its domain is C (community) and P (private).

Establish the following constraints:

2. The *price* field must be equal to or greater than 100.000€.

3. The *register date* must be later than year 2000.

4. A visit cannot be inserted unless the hour ofvisit is between 9 and 15.

5. Remove the previous constraint about the *visit date*.

6. Delete the *email* column of the CLIENT table.

7. Two clients cannot have the same identification card.

8. Change the VISIT table primary key for the *visit date* and *client* fields.

9. We want to ensure that every visit has only ONE client (that is, two clients can visit the same property only at different time). Create a unique index to validate that.

10.Rename CLIENT table for BUYER.

11. Remove the VISIT table.

12.Create a user with your name and a password. Give SELECT and INSERT privileges on the PROPERTY table to that user.

```
USE ESTATE_AGENCY;

#Create a pool field in the PROPERTY table. Its domain is C(community) and P(private).#
#The price field must be equal to or greater than 100.000€#
ALTER TABLE PROPERTY ADD
(POOL CHAR (1) NOT NULL, CHECK ('C' OR 'P'),
CONSTRAINT CHK_PRICE CHECK (PRICE >='100000'));

#The register date must be later than year 2000#
ALTER TABLE CLIENTS ADD
(CONSTRAINT CHK_REGISTER_DATE CHECK (REGISTER_DATE >= '2000-01-01'));

#A visit cannot be inserted unless the hour of visit is between 9 and 15#
ALTER TABLE VISIT ADD
(CONSTRAINT CHK_VISIT_DATE_HOUR CHECK(VISIT_DATE_HOUR BETWEEN '9:00' AND '15:00'));

#Remove the previous constraint about the visit date#
ALTER TABLE VISIT DROP CONSTRAINT CHK_VISIT_DATE_HOUR;

#Delete the email column of the CLIENT table#
ALTER TABLE CLIENTS DROP COLUMN EMAIL;

#Two clients cannot have the same identification card#
ALTER TABLE CLIENTS ADD
(CONSTRAINT ID_CLIENT UNIQUE (ID_CLIENT));

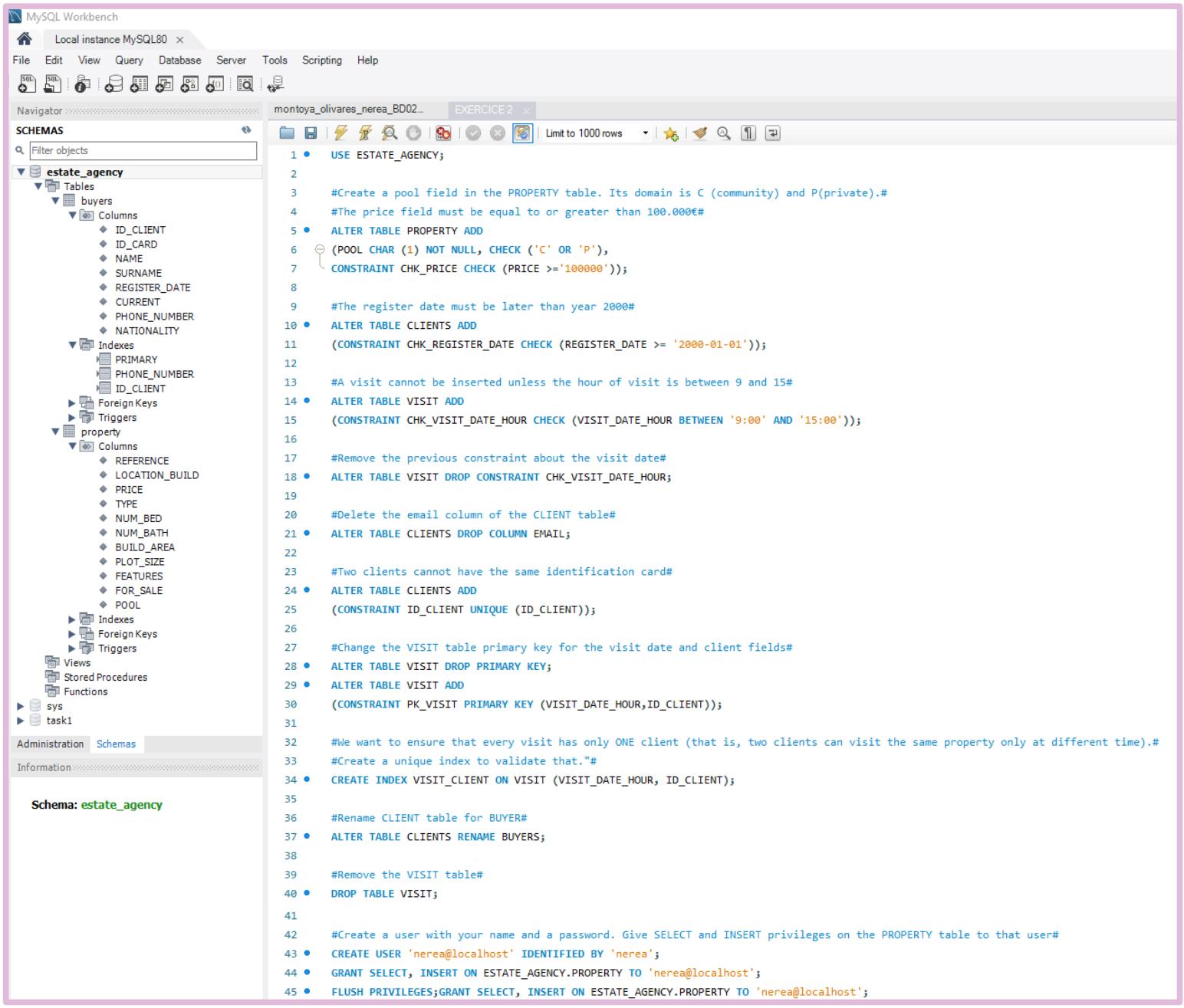
#Change the VISIT table primary key for the visit date and client fields#
ALTER TABLE VISIT DROP PRIMARY KEY;
ALTER TABLE VISIT ADD
(CONSTRAINT PK_VISIT PRIMARY KEY (VISIT_DATE_HOUR,ID_CLIENT));

#We want to ensure that every visit has only ONE client#
#(two clients can visit the same property only at different time).#
#Create a unique index to validate that."#
CREATE INDEX VISIT_CLIENT ON VISIT (VISIT_DATE_HOUR, ID_CLIENT);

#Rename CLIENT table for BUYER#
ALTER TABLE CLIENTS RENAME BUYERS;

#Remove the VISIT table#
DROP TABLE VISIT;

#Create a user with your name and a password. Give SELECT and INSERT privileges on the
PROPERTY table to that user#
CREATE USER 'nerea@localhost' IDENTIFIED BY 'nerea';
GRANT SELECT, INSERT ON ESTATE_AGENCY.PROPERTY TO 'nerea@localhost';
FLUSH PRIVILEGES;GRANT SELECT, INSERT ON ESTATE_AGENCY.PROPERTY TO
'nerea@localhost';
```


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
estate_agency
Tables
buyers
Columns
ID_CLIENT
ID_CARD
NAME
SURNAME
REGISTER_DATE
CURRENT
PHONE_NUMBER
NATIONALITY
Indexes
PRIMARY
PHONE_NUMBER
ID_CLIENT
Foreign Keys
Triggers
property
Columns
REFERENCE
LOCATION_BUILD
PRICE
TYPE
NUM_BED
NUM_BATH
BUILD_AREA
PLOT_SIZE
FEATURES
FOR_SALE
POOL
Indexes
Foreign Keys
Triggers
Views
Stored Procedures
Functions
sys
task1
Administration Schemas
Information
Schema: estate_agency
montoya_olivares_nerea_BD02...
EXERCICE 2
Limit to 1000 rows
1 USE ESTATE_AGENCY;
2
3 #Create a pool field in the PROPERTY table. Its domain is C (community) and P(private).#
4 #The price field must be equal to or greater than 100.000€#
5 ALTER TABLE PROPERTY ADD
6 (POOL CHAR (1) NOT NULL, CHECK ('C' OR 'P'),
7 CONSTRAINT CHK_PRICE CHECK (PRICE >='100000'));
8
9 #The register date must be later than year 2000#
10 ALTER TABLE CLIENTS ADD
11 (CONSTRAINT CHK_REGISTER_DATE CHECK (REGISTER_DATE >= '2000-01-01'));
12
13 #A visit cannot be inserted unless the hour of visit is between 9 and 15#
14 ALTER TABLE VISIT ADD
15 (CONSTRAINT CHK_VISIT_DATE_HOUR CHECK (VISIT_DATE_HOUR BETWEEN '9:00' AND '15:00'));
16
17 #Remove the previous constraint about the visit date#
18 ALTER TABLE VISIT DROP CONSTRAINT CHK_VISIT_DATE_HOUR;
19
20 #Delete the email column of the CLIENT table#
21 ALTER TABLE CLIENTS DROP COLUMN EMAIL;
22
23 #Two clients cannot have the same identification card#
24 ALTER TABLE CLIENTS ADD
25 (CONSTRAINT ID_CLIENT UNIQUE (ID_CLIENT));
26
27 #Change the VISIT table primary key for the visit date and client fields#
28 ALTER TABLE VISIT DROP PRIMARY KEY;
29 ALTER TABLE VISIT ADD
30 (CONSTRAINT PK_VISIT PRIMARY KEY (VISIT_DATE_HOUR,ID_CLIENT));
31
32 #We want to ensure that every visit has only ONE client (that is, two clients can visit the same property only at different time).#
33 #Create a unique index to validate that."#
34 CREATE INDEX VISIT_CLIENT ON VISIT (VISIT_DATE_HOUR, ID_CLIENT);
35
36 #Rename CLIENT table for BUYER#
37 ALTER TABLE CLIENTS RENAME BUYERS;
38
39 #Remove the VISIT table#
40 DROP TABLE VISIT;
41
42 #Create a user with your name and a password. Give SELECT and INSERT privileges on the PROPERTY table to that user#
43 CREATE USER 'nerea@localhost' IDENTIFIED BY 'nerea';
44 GRANT SELECT, INSERT ON ESTATE_AGENCY.PROPERTY TO 'nerea@localhost';
45 FLUSH PRIVILEGES;GRANT SELECT, INSERT ON ESTATE_AGENCY.PROPERTY TO 'nerea@localhost';